Cryptography Assignment-2

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Q.1 ⇒ $\cancel{\quad}^{n} = 17$ $(n=17)$

$\phi(17) = (17-1) = 16$

$Z_{17}^{*} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$

⇒ $\cancel{\phi(\quad)}^{42} = 2 \times 3 \times 7$ $(n=42)$

$\phi(42) = n \times \left(1-\frac{1}{P_1}\right)\left(1-\frac{1}{P_2}\right)\left(1-\frac{1}{P_3}\right)$

$\phi(42) = 42 \times \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{7}\right)$

$\phi(42) = \cancel{42}^{\cancel{6}\,\cancel{3}} \times \frac{1}{\cancel{2}} \times \frac{2}{\cancel{3}} \times \frac{6}{\cancel{7}} \Rightarrow 12$

$\boxed{\phi(42) = 12}$ Ans

$Z_{42}^{*} = \{1, 5, 11, 13, 17, 19, 23, 25, 29, 31, 37, 41\}$

⇒ $\phi(n) = n^{e} - n^{e-1}$ $(n=25,\ e=2 \text{ as } 25=5^2)$

$\phi(25) = 5^2 - 5^1 = 20$

$Z_{25}^{*} = \{1, 2, 3, 4, 6, 7, 8, 9, \cancel{10}, 11, 12, 13, 14, \cancel{15}, 16, 17, 18, 19, \cancel{20}\ 21, 22, 23, 24\}$

Q. 2 ⇒ (a) Additive inverse of 27 $= -27$

under modulo 100:- $-27 \bmod 100$

$\Rightarrow 100 - 27 \Rightarrow 73$

Multiplicative inverse of 27 (mod 100)

$\Rightarrow 63$

(b) $\phi(101) = 100$

$\phi(102) = 102 \times \left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{17}\right)$

$\rightarrow (102)\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{16}{17}\right) \Rightarrow 32$ Ans

$\Rightarrow \phi(500) = 500\left(1-\frac{1}{2}\right)\left(1-\frac{1}{5}\right)$

$\Rightarrow 500\left(\frac{1}{2}\right)\left(\frac{4}{5}\right) \Rightarrow 200$ Ans

Q. 3 $\Rightarrow$ $Z_9^* = 9$ (not prime)

$9 = 3^2$

$\phi(9) = 3^2 - 3^{2-1} \Rightarrow 9-3 = 6$

$\rightarrow$ Therefore 6 elements have their multiplicative inverse under modulo $9 \Rightarrow 1, 2, 4, 5, 7, 8$

$\Rightarrow$ Their inverses are :-

1. $1^{-1} \pmod 9 = 1$
2. $2^{-1} \pmod 9 = 5$
3. $4^{-1} \pmod 9 = 7$
4. $7^{-1} \pmod 9 = 4$
5. $5^{-1} \pmod 8 = 2$
6. $8^{-1} \pmod 9 = 8$

Q. 4 $\Rightarrow$ (a) $6^{-1} \pmod 7$ ($7 \rightarrow$ prime)

$\rightarrow$ using Fermat little's theorem :-

$\Rightarrow 6^{-1} \pmod 7 = 6^{7-2} \bmod 7 \Rightarrow 6^5 \bmod 7$

$\Rightarrow 6$

(b) $a = 7, n = 15$ (not prime)

$a^{\phi(n)} \equiv 1 \bmod n$

$\phi(n) = \phi(15) = \phi(3) \times \phi(5) \Rightarrow 2 \times 4 \Rightarrow 8$

$7^8 \equiv 1 \pmod{15}$

$\therefore$ $a^{-1} \bmod n = a^{\phi(n)-1} \bmod n$

$7^{-1} \bmod 15 \Rightarrow 7^{7} \bmod 15 \Rightarrow 13$

(c) $a = 19$, $n = 101$

$\phi(101) = 101 - 1 = 100$

$\Rightarrow 19^{-1} \bmod 101 \Rightarrow 19^{100-1} \bmod 101$

$\Rightarrow 19^{99} \bmod 101 \Rightarrow 16$ (use fast exp: or extended euclid) (in case of direct inverse).

(d) $a = 97$, $n = 100$

$\rightarrow a^{-1} \bmod n \rightarrow a^{\phi(n)-1} \bmod n$ [Since a & n are co-prime]

$\Rightarrow \phi(100) \Rightarrow 100\left(1-\frac{1}{2}\right)\left(1-\frac{1}{5}\right)$

$\Rightarrow 100\left(\frac{1}{2}\right)\left(\frac{4}{5}\right) \Rightarrow 40$

$\therefore$ $97^{-1} \bmod 100 \rightarrow 97^{39} \bmod 100$

$\Rightarrow (\underbrace{97^2 \times 97^2 \cdots\cdots 97^2}_{\text{(19 times)}} \times 97) \bmod 100$

$\Rightarrow [\underbrace{[97^2 \bmod 100 \times 97^2 \bmod 100 \cdots]}_{\text{18 times}} \times 97 \bmod 100] \bmod 100$

$\rightarrow [9^{18} \times 97] \bmod 100$

$\rightarrow (9^{18} \bmod 100 \times 97 \bmod 100) \bmod 100$

$\Rightarrow [[[9^9 \bmod 100 \times 9^9 \bmod 100] \bmod 100] \times 97 \bmod 100] \bmod 100$

$\rightarrow (((89 \times 89) \bmod 100) \times 97) \bmod 100) \bmod 100 \rightarrow \cancel{37}$ Ans

Q.5 $\Rightarrow$ (a) $x \equiv 4^{100} \bmod 7$

$x = (4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10} \times 4^{10}) \bmod 7$

$x = 4^{10} \bmod 7 \Rightarrow 1048576 \bmod 7 \Rightarrow 4$

(b) $9^x \equiv 13 \pmod{17}$

$\because 13 \equiv 81 \pmod{17}$

$\therefore 9^x = 81 \Longrightarrow \boxed{x = 2}$ Ans

(c) $5^x \equiv 13 \pmod{17}$

$\because 625 \equiv 13 \pmod{17}$; $5^x = 625$; $\boxed{x = 4}$ Ans

(d) $7^x \equiv 11 \pmod{13}$

$\because 7^5 \equiv 11 \pmod{13}$ $\therefore \boxed{x = 5}$ Ans

Q.7 $\Rightarrow$ (a) 726 and 1144 :-

q	r_1	r_2	r
1	1144	726	418
1	726	418	308
1	418	308	110
2	308	110	88
1	110	88	22
4	88	22	0
	22	0	

$\downarrow$

GCD

$\therefore$ GCD(726, 1144) $\Rightarrow$ 22 Ans

(b) 2184 & 16170

q	r_1	r_2	r
7	16170	2184	882
2	2184	882	420
2	882	420	42
10	420	42	0
	42	0	

$\therefore$ GCD(2184, 16170) $\Rightarrow$ 42 Ans

Q. 8 ⇒ Extended Euclidean Algorithm :-

$654\,u + 123\,v = GCD(654, 123)$

$s \cdot u + t \cdot v = GCD(654, 123)$

→ finding s & t using extended Euclidean Algo :-

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	654	123	39	1	0	1	0	1	-5
3	123	39	6	0	1	-3	1	-5	16
6	39	6	3	1	-3	19	-5	16	-101
2	6	3	0	-3	19	-41	16	-101	218
	3	0		19	-41		-101	218	
	↓ GCD			↓ s			↓ t		

⇒ $s \cdot u + t \cdot v = GCD(654, 123)$

⇒ $654 \times (19) + 123 \times (-101) = GCD(654, 123)$

⇒ $12426 - 12423 \Rightarrow 3 \Rightarrow GCD(654, 123)$

Q. 9 ⇒ $13^{-1} \bmod 31$

→ using extended Euclidean :-

q	r_1	r_2	r	t_1	t_2	t
2	31	13	5	0	1	-2
2	13	5	3	1	-2	5
1	5	3	2	-2	5	-7
1	3	2	1	5	-7	12
2	2	1	0	-7	12	-31
	1	0		12	-31	
	↓ GCD			↓ t		

∴ $13^{-1} \bmod 31 \Rightarrow 12$ Ans

Q.10 ⇒ (a) $y^2 \equiv 3 \mod 143$

$289 \equiv 3 \pmod{143}$; $17^2 \equiv 3 \pmod{143}$

$\therefore \boxed{y = 17}$ Ans

(b) $y^2 \equiv 421 \pmod{693}$

$2500 \equiv 421 \pmod{693}$

$y^2 = 2500$; $\boxed{y = 50}$ Ans